

n -SCRAMBLED CANTOR SETS

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ABSTRACT. For all $n \geq 2$, we provide an example of a homeomorphism of Cantor space that has a distributionally n -scrambled Cantor set containing a distributionally $(n+1)$ -scrambled uncountable set, but does not have an $(n+1)$ -scrambled Cantor set.

INTRODUCTION

Given $n \geq 2$, a metric space X , $f: X \rightarrow X$, and $x \in X^n$, define $\underline{d}_X(x) = \min_{j < k < n} d_X(x(j), x(k))$ and $\bar{d}_X(x) = \max_{j < k < n} d_X(x(j), x(k))$. We say that x is *proximal* if $\liminf_{i \rightarrow \infty} \bar{d}_X(f^i \circ x) = 0$, *distributionally proximal* if $\forall \epsilon > 0 \limsup_{i \rightarrow \infty} |\{i' < i \mid \bar{d}_X(f^{i'} \circ x) < \epsilon\}|/i = 1$, *separated* if $\limsup_{i \rightarrow \infty} \underline{d}_X(f^i \circ x) > 0$, *distributionally separated* if $\exists \epsilon > 0 \liminf_{i \rightarrow \infty} |\{i' < i \mid \underline{d}_X(f^{i'} \circ x) < \epsilon\}|/i = 0$, and *(distributionally) scrambled* if it is (distributionally) proximal and (distributionally) separated. A set $Y \subseteq X$ is *(distributionally) n -scrambled* if every injection in Y^n is (distributionally) scrambled, and *distributionally finitely scrambled* if it is distributionally n -scrambled for all $n \geq 2$.

In [GGM21], it was shown that 2-scrambled uncountable sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show that the analogous statement for larger n is false:

Theorem 1 (ZFC). *Suppose that $n \geq 2$. Then there is a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ that has a distributionally n -scrambled Cantor set containing a distributionally finitely scrambled uncountable set, yet does not have an $(n+1)$ -scrambled Cantor set.*

Let INJ_X^n denote the set of injective sequences of elements of X of length n . Let $(\text{D})\text{SCR}_f$ denote the set of (distributionally) scrambled finite sequences. A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. We say that Y is a *G -clique* if its distinct points are G -related, and *G -independent* if its points are not G -related. Let $\text{CLQ}_G^{\geq n}$ denote the set of injective finite sequences of length at least n whose images are G -cliques, and let IND_G^n denote the set of injective sequences of

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length n whose images are G -independent. A *homomorphism* from a relation R on X to a relation S on Y is a function $\pi: X \rightarrow Y$ whose composition with each sequence in R is in S . Such a function is a *homomorphism* from (R, R') to (S, S') if it is also a homomorphism from R' to S' . The *saturation* of Y under a bijection $T: X \rightarrow X$ is given by $[Y]_T = \bigcup_{i \in \mathbb{Z}} T^i(Y)$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is as follows:

Theorem 2 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$ and $n \geq 2$. Then there exist a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ from $(\text{INJ}_{2^\mathbb{N}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{DSCR}_T, \sim \text{SCR}_T)$ for which $[\pi(2^\mathbb{N})]_T$ is co-countable.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

1. HYPERGRAPH HOMOMORPHISMS

Define $S: 2^\mathbb{Z} \rightarrow 2^\mathbb{Z}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^\mathbb{Z}$. Let $(\text{D})\text{SEP}_f$ denote the set of (distributionally) separated finite sequences. As (distributional) scrambling is preserved by topological conjugacy of compact metric spaces, Theorem 2 is a consequence of the topological characterization of $2^\mathbb{N}$ (see, for example, [Kec95, Proposition 7.4]) and:

Theorem 1.1 (ZF + DC). *Suppose that G is an F_σ graph on $2^\mathbb{N}$ and $n \geq 2$. Then there exist an S -invariant perfect set $F \subseteq 2^\mathbb{Z}$ and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow F$ from $(\text{INJ}_{2^\mathbb{N}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{DSCR}_S, \sim \text{SEP}_S)$ for which $F \setminus [\pi(2^\mathbb{N})]_S$ is countable.*

Proof. For all $s \in 2^{<\mathbb{N}}$, set $\mathcal{N}_s = \{c \in 2^\mathbb{N} \mid \forall i < |s| \ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, set $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $\ell \in \mathbb{N}$, $n' \geq n$, and $p: 2^\ell \rightarrow 2^{n'}$, set $\mathcal{N}_p = \prod_{i < n'} \mathcal{N}_{p^{-1}(\{i\})}$. We say that p is a *near bijection* if it induces a bijection of $2^\ell \setminus p^{-1}(\{0\})$ with $n' \setminus \{0\}$ but $\{0^\ell\} \subsetneq p^{-1}(\{0\})$. For all $0 < k < n'$, let $p^{-1}(k)$ denote the unique element of $p^{-1}(\{k\})$.

Fix an increasing sequence $(G_m)_{m \in \mathbb{N}}$ of closed graphs on $2^\mathbb{N}$ whose union is G . Define $G_{\ell, m} = \{(c, c') \in 2^\mathbb{N} \times 2^\mathbb{N} \mid G_m \cap (\mathcal{N}_{c|\ell} \times \mathcal{N}_{c'|\ell}) \neq \emptyset\}$. Let P_ℓ denote the set of near bijections $p: 2^\ell \rightarrow n$, and let $P_{\ell, m, n'}$ denote the set of near bijections $p: 2^\ell \rightarrow n'$ such that $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{\ell, m}$ for all $0 < j < k < n'$. Fix an infinite set $I \subseteq \mathbb{N} \setminus \{0\}$, a bijection $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{\ell \in \mathbb{N}} (\{\ell\} \times P_\ell) \cup \bigcup_{\ell, m \in \mathbb{N}, n' > n} (\{\ell\} \times \{m\} \times \{n'\} \times P_{\ell, m, n'})$, and a function $r: I \rightarrow \mathbb{N} \setminus \{0\}$ with the property that $i^2 \leq r(i)$ and $I \cap [i, i(i + r(i)q(i))] = \{i\}$ for all $i \in I$, where $w(i) = 0$ if $\phi(i) \in \mathbb{N}$, $w(i) = n - 2$ if $\phi(i) \in \{\ell\} \times P_\ell$ for some $\ell \in \mathbb{N}$, $w(i) = (m + 1)(n' - 2)$

if $\phi(i) \in \{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'}$ for some $\ell, m \in \mathbb{N}$ and $n' > n$, and $q(i) = \max(1, 2w(i))$. Define a continuous function $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{Z}}$ by setting $\pi(c)(i') = 1$ if and only if either $i' = 0$ or there exist $i \in I$ and $t < r(i)$ such that one of the following holds:

- (1) $\phi(i) \in \mathbb{N}$, $i' = i + tq(i)$, and $c(\phi(i)) = 1$;
- (2) $\exists 0 < k < n \exists \ell \in \mathbb{N} \exists p \in P_{\ell}$
 $(\ell, p) = \phi(i)$, $i' = i + tq(i) + k - 1$, and $p(c \upharpoonright \ell) = k$; or
- (3) $\exists n' > n \exists 0 < k < n' \exists \ell, m \in \mathbb{N} \exists p \in P_{\ell,m,n'}$
 $(\ell, m, n', p) = \phi(i)$, $i' = i + tq(i) + (k - 1)(m + 1)$, and
 $p(c \upharpoonright \ell) = k$.

Lemma 1.2. π is injective.

Proof. Suppose that $c, d \in 2^{\mathbb{N}}$ are distinct. Then there exists $k \in \mathbb{N}$ with $c(k) \neq d(k)$. Fix $i \in I$ for which $\phi(i) = k$. As $\pi(c)(i) = c(k)$ and $\pi(d)(i) = d(k)$, it follows that $\pi(c)(i) \neq \pi(d)(i)$, so $\pi(c) \neq \pi(d)$. $\square$

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c, d) = \sum_{i \in \mathbb{Z}} |c(i) - d(i)|/2^{|i|}$ for all $c, d \in 2^{\mathbb{Z}}$. For all $i \in I$, set $N_i = [i, i + w(i)]$ and $N'_i = \bigcup_{t < r(i)} N_{i,t}$, where $N_{i,t} = N_i + tq(i)$.

Lemma 1.3. π is a homomorphism from $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$ to DSEP_S .

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n'}$ is in $\text{CLQ}_G^{\geq n+1} \cup \text{INJ}_{2^{\mathbb{N}}}^n$. By permuting the coordinates of c , we can assume that $c(0) = 0^{\mathbb{N}}$ if $0^{\mathbb{N}} \in c(n')$. If $n' > n$, then fix $m \in \mathbb{N}$ such that $c(j) \leq G_m c(k)$ for all $j < k < n'$. Suppose now that $\ell \in \mathbb{N}$ is sufficiently large that $\text{proj}_{2^{\ell}} \circ c$ is injective, $c(k) \upharpoonright \ell \neq 0^{\ell}$ for all $0 < k < n'$, and $n' < 2^{\ell}$. If $n' = n$, then let p be the extension of $(\text{proj}_{2^{\ell}} \circ c)^{-1}$ in P_{ℓ} , and fix $i \in I$ such that $\phi(i) = (\ell, p)$. If $n' > n$, then let p be the extension of $(\text{proj}_{2^{\ell}} \circ c)^{-1}$ in $P_{\ell,m,n'}$, and fix $i \in I$ such that $\phi(i) = (\ell, m, n', p)$. If $t < r(i)$, then $\text{proj}_{2^{N_{i,t}}} \circ \pi \circ c$ is injective, so $\rho(S^{i'} \circ \pi \circ c) \geq 1/2^{q(i)}$ for all $i \leq i' < i + q(i)r(i)$, thus

$$\begin{aligned} & |\{i' < i + q(i)r(i) \mid \rho(S^{i'} \circ \pi \circ c) < 1/2^{q(i)}\}| / (i + q(i)r(i)) \\ & \leq i / (i + q(i)r(i)) \leq 1 / (1 + iq(i)) \leq 1 / (1 + i) \rightarrow \infty \end{aligned}$$

as $\ell \rightarrow \infty$, hence $\pi \circ c \in \text{DSEP}_S$. $\square$

Lemma 1.4. π is a homomorphism from IND_G^{n+1} to $\sim\text{SEP}_S$.

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n+1}$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \rho(S^{i'} \circ \pi \circ c) \geq \epsilon\}$ is infinite, as well as $a \in \mathbb{N}$ such that $\sum_{|i| > a} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I' , we can assume that, for all $i' \in I'$, there is at most one $i \in I$ such that $[i' - a, i' + a] \cap N'_i \neq \emptyset$, so our choice of a yields a unique such i , in which

case $\text{proj}_{2^{N_{i,t}}} \circ \pi \circ c$ is injective for all $t < r(i)$. It follows that $\phi(i)$ is of the form (ℓ, m, n', p) for some $n' > n$, where $(m+1)(n-1) \leq 2a$ and there exist $j < k < n+1$ with $c(j) G_{\ell,m} c(k)$.

By passing to an infinite subset of I' , we can assume that the values of j, k , and m are independent of $i' \in I'$. Then there are infinitely many values of ℓ corresponding to the remaining $i' \in I'$, since otherwise there are only finitely many possibilities for both n' and p , and therefore only finitely many for i , which contradicts the fact that any two elements of I' corresponding to the same element of I must be within $2a$ of one another. So $c(j) G_m c(k)$, thus $c(j) G c(k)$, hence $c \notin \text{IND}_G^{n+1}$. $\square$

Clearly, the closure F of $[\pi(2^{\mathbb{N}})]_S$ is perfect.

Lemma 1.5. *$F \setminus [\pi(2^{\mathbb{N}})]_S$ is countable.*

Proof. If $(c_k)_{k \in \mathbb{N}}$ is a sequence of elements of $2^{\mathbb{N}}$ and $(i_k)_{k \in \mathbb{N}}$ is a strictly increasing sequence of elements of $\mathbb{N}$, then $S^{-i_k}(\pi(c_k)) \rightarrow 0^{\mathbb{Z}}$. Suppose that $S^{i_k}(\pi(c_k)) \rightarrow c$. If the distance from i_k to $\bigcup_{i \in I} N'_i$ goes to infinity, then $c = 0^{\mathbb{Z}}$, so we can assume that there exist $i'_k \in I$ such that the distance from i_k to $N'_{i'_k}$ is fixed.

If $w(i'_k) \rightarrow \infty$, then every fixed neighborhood of i_k eventually meets at most one $N'_{i'_k}$, so the support of c has cardinality at most one. Otherwise, by passing to a subsequence, we can assume that $w(i'_k)$ is constant. On $[i'_k, i'_k + q(i'_k)r(i'_k))$, each $\pi(c_k)$ is either zero or the restriction of a periodic sequence of period $q(i'_k)$. Since $r(i'_k) \rightarrow \infty$, the sequence c is either periodic or obtained from a periodic sequence by replacing all coordinates on one side of an integer by zero. $\square$

To see that each $c \in \pi(2^{\mathbb{N}})^{<\mathbb{N}}$ is distributionally proximal, suppose that $\epsilon > 0$ and fix $a \in \mathbb{N}$ with $\sum_{|k|>a} 1/2^{|k|} < \epsilon$. Given $i \in I$, let i' be the next element of I . If $i + q(i)r(i) + a \leq i'' < i' - a$, then $\bigcup_{k < |c|} \text{supp}(c(k)) \subseteq \{0\} \cup \bigcup_{i \in I} N'_i$, so $\bar{\rho}(S^{i''} \circ c) < \epsilon$, thus

$$\begin{aligned} |\{i'' < i' \mid \bar{\rho}(S^{i''} \circ c) < \epsilon\}|/i' &\geq 1 - (i + q(i)r(i) + 2a)/i' \\ &\geq 1 - (1/i) - (2a/i') \rightarrow 1 \end{aligned}$$

as $i \rightarrow \infty$. $\square$

Proposition 1.6 (ZF + DC). *Suppose that $n \geq 3$, X and Y are Polish spaces, G is a Borel graph on X , $T: Y \rightarrow Y$ is a Borel automorphism, $B \subseteq Y$ is an n -scrambled uncountable Borel set, $\pi: X \rightarrow Y$ is an injective Borel homomorphism from IND_G^n to $\sim \text{SCR}_T$, and $[\pi(X)]_T$ is co-countable. Then G has a perfect clique.*

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)(X)$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G -clique or G -independent. But G -independence would contradict the fact that B is n -scrambled. $\square$

2. n -SCRAMBLED SETS

We begin this section with the proof of our primary result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a simple Baire category argument, there is an F_σ graph G on $2^\mathbb{N}$ with an uncountable clique Y but no perfect clique. By Theorem 2, there exist a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ and a continuous injective homomorphism $\pi: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ from $(\text{INJ}_{2^\mathbb{N}}^n \cup \text{CLQ}_G^{\geq n+1}, \text{IND}_G^{n+1})$ to $(\text{DSCR}_T, \sim \text{SCR}_T)$ for which $[\pi(2^\mathbb{N})]_T$ is co-countable. Then $\pi(2^\mathbb{N})$ is distributionally n -scrambled, $\pi(Y)$ is a distributionally finitely scrambled uncountable set, and Proposition 1.6 rules out an $(n+1)$ -scrambled Cantor set. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 (ZFC + $\neg\text{CH}$). *It is consistent that, for all $n \geq 2$, there is a homeomorphism $T: 2^\mathbb{N} \rightarrow 2^\mathbb{N}$ that has a distributionally n -scrambled Cantor set containing a distributionally finitely scrambled set of cardinality $\mathfrak{c}$, but does not have an $(n+1)$ -scrambled Cantor set.*

Proof. By [She99, Theorem 1.13], we can assume that there is an F_σ graph G on $2^\mathbb{N}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 (ZFC + $\neg\text{CH}$). *It is consistent that, for all $n \geq 2$, every Borel function on a Polish space with a (distributionally) n -scrambled set of cardinality $\aleph_2$ has a (distributionally) n -scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel n -ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But n -ary (distributional) scrambledness is Borel. $\square$

Remark. The proof of [KS03, Proposition 3.4] can be used to establish the analog of Theorem 2.2 for (distributionally) finitely scrambled sets.

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